Discriminative Learning: Linear Classifiers II

Support vector machines and maximum-margin classification

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Learning objectives

By the end of this lecture, you should be able to:

1. Compute distance and margin for linear classifiers.
2. Explain hard and soft margins, support vectors, and the dual.
3. Use hinge loss and interpret the parameter C.

Beyond classification accuracy

Last week, we learned to fit a linear score $f(\boldsymbol{x}) = \boldsymbol{w}^{\mathsf{T}}\boldsymbol{x} + b$.

- Perceptron learning corrects mistakes.
- Logistic regression fits conditional class probabilities.

Both methods can separate training data, but a separable dataset may admit many valid boundaries.

SVM adds a preference: keep the boundary far from the closest points.

We first study perfect separation, then allow violations for noisy data.

Margin geometry

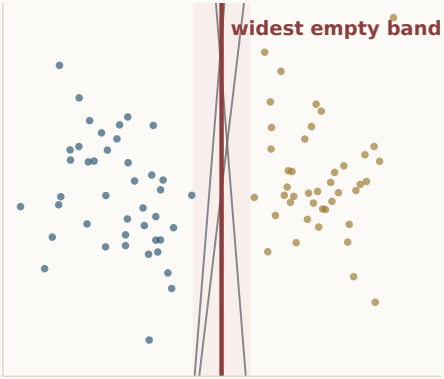
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Many separators

For separable labels $y_i \in \{-1, +1\}$, many linear boundaries satisfy:

$$y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) > 0.$$

All have zero training errors, but their distances to the closest points differ. **SVM:** choose the widest separating margin.



Equal accuracy, different margins

Suppose negative examples are at $-2, -1$ and positive examples are at $2, 3$ on a number line.

Decision boundary	Training errors	Closest-point distance
$x = 0$	0	$\min(1, 2) = 1$
$x = 0.5$	0	$\min(1.5, 1.5) = 1.5$

Both boundaries are perfectly accurate, but $x = 0.5$ stays farther from the nearest training example.

The margin distinguishes separators that have identical training accuracy.

Linear boundary

For features $\boldsymbol{x} \in \mathbb{R}^d$, a linear classifier uses weights $\boldsymbol{w} \in \mathbb{R}^d$ and intercept b:

$$\boxed{f(\boldsymbol{x}) = \boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b}.$$

The boundary is $f(\boldsymbol{x}) = 0$, with $\boldsymbol{w}$ normal to it.

- $f(\boldsymbol{x}) > 0$: predict $+1$.
- $f(\boldsymbol{x}) < 0$: predict -1.

The score gives the side of the boundary; next we turn it into a distance.

Score is not distance

Multiplying a score by a positive constant does not move its boundary:

$$f(\boldsymbol{x}) = \boldsymbol{w}^{\mathsf{T}}\boldsymbol{x} + b, \qquad f'(\boldsymbol{x}) = 2f(\boldsymbol{x}).$$

Both have the same zero set and predictions, but $|f'| = 2|f|$.

Raw score magnitude is not geometric distance.

We must divide by the length of the normal vector.

Point-to-boundary distance

Consider the hyperplane:

$$H: \quad \boldsymbol{w}^{\mathsf{T}}\boldsymbol{x} + b = 0.$$

Its unit normal direction is:

$$\frac{\boldsymbol{w}}{\|\boldsymbol{w}\|_2}.$$

The perpendicular distance from $\boldsymbol{x}$ to H is:

$$\boxed{\operatorname{dist}(\boldsymbol{x}, H) = \frac{|\boldsymbol{w}^{\mathsf{T}}\boldsymbol{x} + b|}{\|\boldsymbol{w}\|_2}}.$$

Dividing by $\|\boldsymbol{w}\|_2$ removes the arbitrary scale of the score.

Distance example

Consider $\boldsymbol{w} = (2, 0)^\mathsf{T}$ and $b = -2$. Their linear decision boundary is:

$$2x_1 - 2 = 0 \quad \Longrightarrow \quad \boxed{x_1 = 1}.$$

For $\boldsymbol{x} = (3, 4)^\mathsf{T}$, compute the score and weight norm:

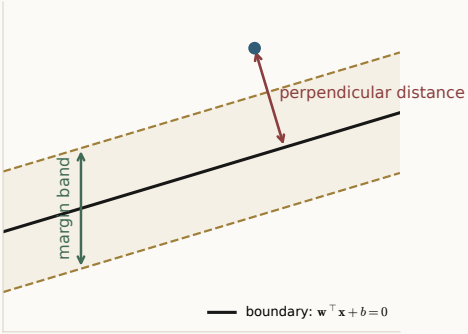
$$f(\boldsymbol{x}) = 2(3) - 2 = 4, \qquad \|\boldsymbol{w}\|_2 = 2.$$

The distance to the boundary is therefore:

$$\boxed{\mathrm{dist}(\boldsymbol{x}, H) = \frac{|4|}{2} = 2.}$$

This agrees with the horizontal distance $|3 - 1| = 2$ to the vertical line $x_1 = 1$.

Distance geometry



The closest observations determine how wide an empty band can be placed around the boundary.

SVM chooses the separator with the widest possible margin band.

Signed margin

For each labeled example $(\boldsymbol{x}_i, y_i)$ with $y_i \in \{-1, +1\}$, define its **functional margin**:

$$\boxed{\hat{\gamma}_i = y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)}.$$

- $\hat{\gamma}_i > 0$: correctly classified.
- $\hat{\gamma}_i < 0$: incorrectly classified.
- Its magnitude depends on the scaling of $(\boldsymbol{w}, b)$.

Next we normalize it to obtain geometric margin.

Geometric margin

Normalize the functional margin by the weight norm:

$$\boxed{\gamma_i = \frac{y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)}{\|\boldsymbol{w}\|_2}}.$$

γ_i is the **signed perpendicular distance** to the boundary.

- Positive for a correctly classified point; negative otherwise.
- Unchanged when $(\boldsymbol{w}, b)$ are multiplied by $c > 0$.

This is the meaningful quantity to maximize.

Signed distance calculation

Take $\boldsymbol{w} = (3, 4)^{\mathsf{T}}$, $b = -5$. Then $\|\boldsymbol{w}\|_2 = 5$ and $f(\boldsymbol{x}) = 3x_1 + 4x_2 - 5$.

Point	Label y	Score $f(\boldsymbol{x})$	Geometric margin $yf/5$
$\boldsymbol{x}_A = (2, 1)^{\mathsf{T}}$	$+1$	5	1
$\boldsymbol{x}_B = (0, 0)^{\mathsf{T}}$	-1	-5	1

Multiplying $(\boldsymbol{w}, b)$ by 3 makes both functional margins 15, but geometric margins remain $15/15 = 1$.

Meaning: the normalized score, not the raw score, measures the distance used in SVM geometry.

Scaling ambiguity

A separator has infinitely many equivalent parameterizations:

$$(\boldsymbol{w}, b), \quad (2\boldsymbol{w}, 2b), \quad (10\boldsymbol{w}, 10b).$$

They have the same boundary and geometric margins, but different functional margins.
Next step: fix this arbitrary scale so that the smallest functional margin equals 1.

Canonical scaling

For separable data, choose a positive rescaling of $(\boldsymbol{w}, b)$ such that:

$$\boxed{\min_i y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) = 1}.$$

Every training observation then satisfies:

$$y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq 1.$$

The reference levels $f(\boldsymbol{x}) = +1$ and $f(\boldsymbol{x}) = -1$ are the **margin hyperplanes**.

- The boundary $f(\boldsymbol{x}) = 0$ and all predictions are unchanged by this rescaling.
- At the hard-margin optimum, points from both classes touch the margin hyperplanes.

Canonical scaling in numbers

Consider $f(\boldsymbol{x}) = 4x_2 - 4$ for two observations:

$$\boldsymbol{x}_- = (0,0)^\mathsf{T},\ y_- = -1; \qquad \boldsymbol{x}_+ = (0,2)^\mathsf{T},\ y_+ = +1.$$

Both functional margins are $y_i f(\boldsymbol{x}_i) = 4$, so the minimum is 4, not 1.
Divide the coefficients by 4:

$$\boxed{\boldsymbol{w} = (0,1)^\mathsf{T}, \quad b = -1, \quad f(\boldsymbol{x}) = x_2 - 1.}$$

Now both functional margins equal 1. The boundary remains $x_2 = 1$, and both points remain one unit away from it.
Use: canonical scaling lets us compare parameter norms across different separating boundaries.

Margin width

Under canonical scaling, the decision boundary and margin hyperplanes are:

$$f(\boldsymbol{x}) = 0, \qquad f(\boldsymbol{x}) = +1, \qquad f(\boldsymbol{x}) = -1.$$

Each margin hyperplane is $1/\|\boldsymbol{w}\|_2$ away from the decision boundary.
Therefore the **full margin width** is:

$$\boxed{\frac{2}{\|\boldsymbol{w}\|_2}}.$$

Maximizing width is equivalent to reducing the weight norm.

A one-dimensional example

Consider four labeled observations on the number line:

$$\text{negative: } 0, 1; \qquad \text{positive: } 3, 4.$$

The two closest opposite-class points are 1 and 3. Placing the boundary halfway between them gives:

$$\boxed{x = 2, \qquad f(x) = x - 2.}$$

The closest points have signed margins $(-1)f(1) = 1$ and $(+1)f(3) = 1$.
The farther points at 0 and 4 both have signed margin 2.
Meaning: the training observations at 1 and 3 constrain how wide the margin can be.

Example margin

Small measurement changes can matter near a boundary.
With negative points $0, 1$ and positive points $3, 4$, compare boundaries $x = 1.1$ and $x = 2$.

Boundary	Original errors	Nearest training-point distance
$x = 1.1$	0	0.1
$x = 2$	0	1

Now suppose the negative point at 1 is measured as 1.2.

- Boundary $x = 1.1$ predicts $+1$: incorrect for this perturbed point.
- Boundary $x = 2$ still predicts -1: unchanged.

Use: a wider margin can tolerate larger small perturbations in this simple geometry; it does not by itself guarantee good generalization.

Hard-margin SVM

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From margin to objective

Under canonical scaling, the full margin width is $2/\|\boldsymbol{w}\|_2$.
The margin becomes wider as $\|\boldsymbol{w}\|_2$ becomes smaller. We therefore minimize:

$$\boxed{\frac{1}{2}\|\boldsymbol{w}\|_2^2}.$$

The square and factor $1/2$ preserve the minimizer and simplify differentiation.
But: the minimization must still enforce correct classification with a unit functional margin.

Hard-margin SVM

The hard-margin SVM solves:

$$\begin{aligned} \min_{\boldsymbol{w},b} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 \\ \text{subject to} \quad & y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq 1, \\ & i = 1, \ldots, n. \end{aligned}$$

The two parts have clear roles:

- Objective: make the margin wide.
- Constraints: classify every training observation correctly with margin at least one.

Hard margin in two features

Consider four labeled observations:

Class	Points	Candidate score
-1	$(0,0)$, $(-1,1)$	$f(\boldsymbol{x}) = x_1 - 1$
$+1$	$(2,0)$, $(3,1)$	$f(\boldsymbol{x}) = x_1 - 1$

For $\boldsymbol{w} = (1,0)^{\mathsf{T}}$, $b = -1$, the signed scores are:

$$\boxed{y_i f(\boldsymbol{x}_i) = (1,2,1,2).}$$

All four constraints hold. The closest two points have signed score 1; the other two have score 2.
The norm is 1, so the full margin width is $2/1 = \boxed{2}$.

Optimality from closest points

For the same dataset, the negative point $(0, 0)$ requires:

$$-b \geq 1 \quad \Longrightarrow \quad b \leq -1.$$

The positive point $(2, 0)$ requires:

$$2w_1 + b \geq 1 \quad \Longrightarrow \quad 2w_1 \geq 1 - b \geq 2.$$

Therefore $w_1 \geq 1$, hence $\|\boldsymbol{w}\|_2 \geq 1$. No feasible classifier can have objective below $1/2$. Our candidate $\boldsymbol{w} = (1, 0)^\mathsf{T}$, $b = -1$ achieves exactly:

$$\boxed{\frac{1}{2}\|\boldsymbol{w}\|_2^2 = \frac{1}{2}.}$$

Meaning: the two closest observations already establish the optimal margin; moving the farther points a little will not change it.

Norm and margin

The hard-margin objective is not an arbitrary regularizer:

$$\frac{1}{2}\|\boldsymbol{w}\|_2^2 \quad \longleftrightarrow \quad \text{full width } \frac{2}{\|\boldsymbol{w}\|_2}.$$

Its derivative is especially simple:

$$\nabla_{\boldsymbol{w}} \left(\frac{1}{2}\|\boldsymbol{w}\|_2^2 \right) = \boldsymbol{w}.$$

Connection: the geometric goal gives the optimization objective; the classification constraints keep points on their correct side.

Hard-margin example

Return to:

$$x_- = 1, \qquad x_+ = 3.$$

Choose:

$$w = 1, \qquad b = -2.$$

For the negative point:

$$(-1)(1-2) = 1.$$

For the positive point:

$$(+1)(3-2) = 1.$$

Both constraints are active:

$$y_i f_i = 1.$$

The points lie exactly on the two margin boundaries.

Support vectors

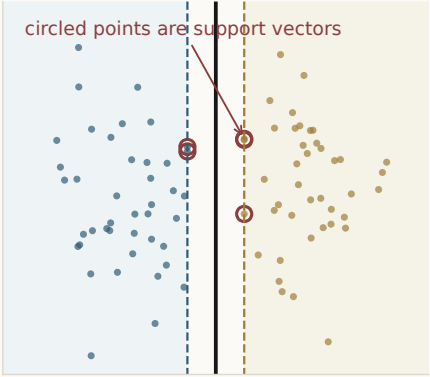
Training observations satisfying:

$$y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) = 1$$

lie on the margin boundary.
They are called:

support vectors.

They are the observations that prevent the margin from becoming wider.



Closest points matter

Suppose a point lies far outside the margin:

$$y_i f_i > 1.$$

Moving it slightly usually does not change the optimal boundary.
But moving a support vector can change:

- the margin width;
- the orientation;
- the intercept.

Thus SVM can depend on only a subset of the training observations.
The dual formulation makes this precise.

Hard-margin limitation

Hard margin requires $y_i f(\boldsymbol{x}_i) \geq 1$ for *every* training observation.

- If classes overlap or a point is mislabeled, the constraints may be impossible to satisfy.
- If one outlier is still separable, it may force a very narrow margin.

We need a way to trade margin width against violations. Before introducing it, the **dual** shows why a small number of training observations can determine the hard-margin separator.
The same dual machinery will later explain the role of C in soft margin.

Dual formulation

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Dual motivation

The primal optimizes $\boldsymbol{w}$ and b. Its constraints, however, are attached to individual training examples.
The dual assigns a nonnegative multiplier α_i to each constraint.

- Identify which points determine the boundary.
- Express the classifier using pairwise inner products.
- Prepare for nonlinear kernels in the next lecture.

First derive the dual for the hard-margin problem.

Constraint form

Rewrite the hard-margin constraints as:

$$\boxed{1 - y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \leq 0}, \qquad i = 1, \ldots, n.$$

Attach one multiplier $\alpha_i \geq 0$ to each constraint.
This allows the constrained problem to be written through a Lagrangian, while the coefficients α_i reveal which constraints are active.

Multipliers and margin constraints

For a signed score $m_i = y_i f(\boldsymbol{x}_i)$, the Lagrangian adds $\alpha_i(1 - m_i)$.
For illustration, take $\alpha_i = 2$:

m_i	$2(1 - m_i)$	Constraint status
1.4	-0.8	Satisfied
1	0	Active
0.7	0.6	Violated

At the optimum, complementary slackness links nonzero multipliers to the margin constraints.

Lagrangian

The goal is still to minimize $\|\boldsymbol{w}\|_2^2/2$ *while enforcing* $1 - y_i f_i \leq 0$.
The Lagrangian combines both:

$$\mathcal{L}(\boldsymbol{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\boldsymbol{w}\|_2^2 + \sum_i \alpha_i \left[1 - y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)\right].$$

The first term rewards a wide margin; each $\alpha_i \geq 0$ is associated with one classification constraint.
Next: minimize this expression over $\boldsymbol{w}$ and b, then maximize the resulting dual objective over admissible $\boldsymbol{\alpha}$.

Remove $\boldsymbol{w}$

Differentiate with respect to $\boldsymbol{w}$:

$$\nabla_{\boldsymbol{w}} \mathcal{L} = \boldsymbol{w} - \sum_i \alpha_i y_i \boldsymbol{x}_i.$$

At the optimum:

$$\nabla_{\boldsymbol{w}} \mathcal{L} = \mathbf{0}.$$

Therefore:

$$\boxed{\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i}.$$

The normal vector is a weighted combination of training observations.

Remove b

Differentiate with respect to b:

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_i \alpha_i y_i.$$

At the optimum:

$$\boxed{\sum_i \alpha_i y_i = 0}.$$

So positive- and negative-class contributions must balance in the multiplier-weighted sum. These two stationary conditions allow us to eliminate $\boldsymbol{w}$ and b.

Substitute $\boldsymbol{w}$

Using:

$$\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i,$$

its squared norm is:

$$\|\boldsymbol{w}\|^2 = \sum_i \sum_j \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^\mathsf{T} \boldsymbol{x}_j.$$

Therefore all interactions between training observations appear through:

$$\boxed{\boldsymbol{x}_i^\mathsf{T} \boldsymbol{x}_j}.$$

These are pairwise inner products.

Dual objective

After substitution, the dual becomes:

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^\mathsf{T} \boldsymbol{x}_j \\ \text{subject to} \quad & \alpha_i \geq 0, \\ & \sum_i \alpha_i y_i = 0. \end{aligned}$$

The optimization variables are now $\alpha_1, \ldots, \alpha_n$.
The training data appear only through pairwise inner products.

Dual optimization in numbers

Consider two training points $x_1 = -1$, $y_1 = -1$ and $x_2 = 1$, $y_2 = +1$.
The dual constraint gives $-\alpha_1 + \alpha_2 = 0$, so let $\alpha_1 = \alpha_2 = \alpha$.
Then the weight is $w = \alpha(-1)(-1) + \alpha(+1)(+1) = 2\alpha$.
The dual objective reduces to:

$$W(\alpha) = 2\alpha - \frac{1}{2}(2\alpha)^2 = 2\alpha - 2\alpha^2.$$

Differentiate: $W'(\alpha) = 2 - 4\alpha = 0$, giving $\boxed{\alpha = 0.5}$ and $\boxed{w = 1}$.
The margin constraints yield $b = 0$, so $\boxed{f(x) = x}$.
Meaning: optimizing the multipliers recovers the maximum-margin classifier without optimizing w directly.

Meaning of α_i

The dual introduces one coefficient $\alpha_i \geq 0$ for each observation:

$$\boxed{\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i}.$$

- If $\alpha_i = 0$, observation i does not enter this expansion.
- If $\alpha_i > 0$, it helps determine the separator.

Next: complementary slackness tells us when a coefficient can be nonzero.

Complementary slackness

At a hard-margin optimum, each multiplier satisfies:

$$\boxed{\alpha_i \left[y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) - 1 \right] = 0.}$$

The product is zero, so:

- If $y_i f_i > 1$ (strictly outside the margin), then $\alpha_i = 0$.
- If $\alpha_i > 0$, then $y_i f_i = 1$ (on the margin).

A point on the margin can have $\alpha_i = 0$ in a degenerate solution; positive multipliers identify the observations that actually appear in the dual prediction.

Support-vector weights connect geometric closeness to the algebra of the learned classifier.

Dual prediction

Substitute $\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i$ into the linear score:

$$\boxed{f(\boldsymbol{x}) = \sum_i \alpha_i y_i \boldsymbol{x}_i^\mathsf{T} \boldsymbol{x} + b}.$$

Only examples with $\alpha_i > 0$ contribute.
The training data appear through inner products $\boldsymbol{x}_i^\mathsf{T} \boldsymbol{x}$: this is the bridge to the kernel method.

A dual example

Consider the two support vectors $x_1 = 1$, $y_1 = -1$ and $x_2 = 3$, $y_2 = +1$.
The dual equality constraint and weight expansion give:

$$-\alpha_1 + \alpha_2 = 0, \qquad w = -\alpha_1(1) + \alpha_2(3).$$

Hence $\alpha_1 = \alpha_2 = \alpha$ and $w = 2\alpha$.
The maximum-margin solution for these points has $w = 1$, so:

$$\boxed{\alpha_1 = \alpha_2 = \tfrac{1}{2}}.$$

Both points have positive multipliers and therefore support the boundary.

Recovering the boundary

For the two support vectors $x_1 = 1$, $y_1 = -1$ and $x_2 = 3$, $y_2 = +1$, the preceding slide gave $w = 1$.
A positive support vector lies on $f(x) = +1$. Substitute $x = 3$:

$$3w + b = 1 \quad \Longrightarrow \quad 3 + b = 1 \quad \Longrightarrow \quad \boxed{b = -2}.$$

Hence the learned maximum-margin score is:

$$\boxed{f(x) = x - 2.}$$

Check the negative support vector: $f(1) = -1$, so $(-1)f(1) = 1$ as required.
Meaning: after the dual finds the weights, any suitable margin support vector can recover the intercept.

An infeasible hard-margin case

Suppose two training records have the *same* features $x = 1$, but opposite labels $+1$ and -1. Hard margin would require both:

$$\underbrace{w(1) + b \geq 1}_{y=+1}, \qquad \underbrace{w(1) + b \leq -1}_{y=-1}.$$

These conditions cannot hold simultaneously, regardless of the chosen w and b.

Why this can happen:

- measurement features may not fully determine the label;
- labels may be noisy or inconsistent.

A practical SVM needs a feasible way to penalize mistakes without forbidding them.

Soft-margin SVM

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Controlled violations

Hard margin can fail when classes overlap or labels conflict. Even a separable outlier may make the margin very narrow.
Soft margin keeps the goal of a wide margin but permits costly violations:

$$\boxed{\text{wide margin} + \text{penalty for violated constraints.}}$$

The next step is to introduce one **slack variable** for each observation.

Slack variables

Give observation i a nonnegative slack variable ξ_i:

$$\boxed{y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0}.$$

ξ_i allows the point to fall short of the desired unit functional margin.
When $y_i f(\boldsymbol{x}_i) \geq 1$, the minimum required slack is 0.
We will penalize larger slacks in the optimization objective.

Repairing infeasible constraints

Return to the two observations at $x = 1$ with labels $+1$ and -1.
Try $w = 0$ and $b = 0$, so both points have signed score $m_i = 0$.
With soft-margin constraints $m_i \geq 1 - \xi_i$, choose:

$$\boxed{\xi_1 = \xi_2 = 1.}$$

Both conditions now hold: $0 \geq 1 - 1 = 0$.

- The model is feasible even though the labels conflict.
- The violations are not free: $C \sum_i \xi_i = 2C$.

Meaning: slack repairs infeasibility but introduces an explicit cost for imperfect separation.

Slack interpretation

Let:

$$m_i = y_i f(\boldsymbol{x}_i).$$

The smallest feasible slack is:

m_i	ξ_i	meaning
$m_i \geq 1$	0	on/outside margin
$0 < m_i < 1$	$1 - m_i$	correct, inside margin
$m_i = 0$	1	on decision boundary
$m_i < 0$	> 1	misclassified

Thus ξ_i measures more than classification error.
It also records insufficient margin.

Slack example

Consider four signed scores $m_i = y_i f(\boldsymbol{x}_i)$:

m_i	1.5	0.6	0	-0.4
Minimum $\xi_i = \max(0, 1 - m_i)$	0	0.4	1	1.4

The correctly classified point with $m_i = 0.6$ still violates the desired margin.
The misclassified point with $m_i = -0.4$ requires the largest slack.

Soft-margin objective

Minimize the weight norm while paying for slack:

$$\begin{aligned} \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_i \xi_i \\ \text{subject to} \quad & y_i(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \qquad C > 0. \end{aligned}$$

Smaller $\|\boldsymbol{w}\|_2$ favors a wider margin; smaller ξ_i favors fewer or milder violations.
C controls the balance.

Role of C

The soft-margin objective balances two terms:

$$\underbrace{\frac{1}{2}\|\boldsymbol{w}\|_2^2}_{\text{margin penalty}} + \underbrace{C\sum_i \xi_i}_{\text{violation penalty}} .$$

- **Large C:** margin violations are expensive; training fit is emphasized.
- **Small C:** violations are less expensive; the optimizer can accept more of them to achieve a smaller weight norm.

This trade-off does not tell us which value gives the best *test* accuracy. Choose C using validation data.

A numerical margin trade-off

Use three labeled records: $(0, -1)$, $(2, +1)$, and $(2, -1)$, where each pair is (x, y).
The last two records conflict. Compare two *candidate* classifiers:

Classifier	$\frac{1}{2}w^2$	Total minimum slack $\sum_i \xi_i$
A: $f(x) = x - 1$	0.5	$0 + 0 + 2 = 2$
B: $f(x) = 0$	0	$1 + 1 + 1 = 3$

Thus $J_A = 0.5 + 2C$ and $J_B = 3C$.
For $C = 0.1$, $J_A = 0.7 > J_B = 0.3$; for $C = 1$, $J_A = 2.5 < J_B = 3$.
Meaning: the preferred *candidate* changes with C. The full SVM optimization may consider other candidates as well.

Hard vs. soft margin

	Hard margin	Soft margin
Constraint violations	Forbidden	Allowed at a cost
Overlapping classes	Infeasible in general	Feasible
Outlier effect	Can dominate the boundary	Controlled by C

Hard margin establishes the geometry; soft margin handles realistic noisy training data.

From slack to loss

For fixed $(\boldsymbol{w}, b)$, the slack constraints imply:

$$\xi_i \geq 1 - y_i f_i, \qquad \xi_i \geq 0.$$

Since the objective penalizes ξ_i with $C > 0$, its optimal value is:

$$\boxed{\xi_i = \max(0, 1 - y_i f_i)}.$$

This is the **hinge loss**: it turns a constrained problem into regularized loss minimization.

Hinge loss in numbers

Hinge loss uses $\max(0, 1 - m)$, where $m = yf(\boldsymbol{x})$ is the signed score.

m	Classification	Perceptron loss	Hinge loss
-0.5	Wrong	0.5	1.5
0.4	Correct, close	0	0.6
1.5	Correct, far	0	0

Key difference: hinge loss still penalizes the correctly classified point with $m = 0.4$ because its margin is too small.
Use: replacing the slack variables by hinge loss turns the constrained problem into regularized empirical-risk minimization.

Hinge loss

Define signed score:

$$m_i = y_i f_i.$$

The hinge loss is:

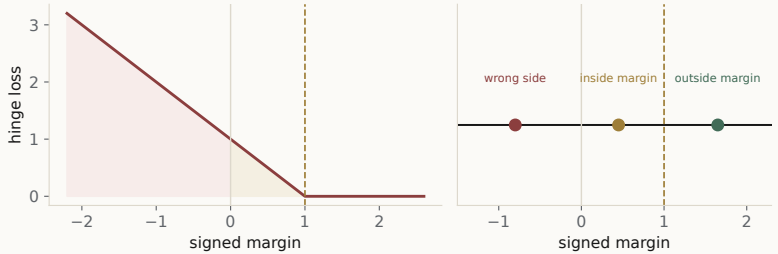
$$\boxed{\ell_{\text{hinge}}(m_i) = \max(0, 1 - m_i)}.$$

Therefore the soft-margin objective becomes:

$$\boxed{\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_i \max(0, 1 - y_i f_i)}.$$

The constrained SVM is now written as regularized loss minimization.

Hinge regimes



- $m_i > 1$: no margin violation, so hinge loss is 0.
- $0 < m_i < 1$: correct class, but the margin is too small.
- $m_i < 0$: incorrect class, with hinge loss greater than 1.

Unlike perceptron loss, hinge loss also penalizes correctly classified points that lie too close to the boundary.

Hinge update

Write the unconstrained soft-margin objective as:

$$J(\boldsymbol{w}, b) = \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_i \max(0, 1 - y_i f_i), \qquad f_i = \boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b.$$

For a point with $m_i = y_i f_i < 1$, its hinge-loss gradient is $-y_i \boldsymbol{x}_i$; for $m_i > 1$, it is zero. Away from the corners $m_i = 1$, the total gradient is:

$$\boxed{\nabla_{\boldsymbol{w}} J = \boldsymbol{w} - C\sum_{i:m_i<1} y_i \boldsymbol{x}_i.}$$

The first term comes from margin regularization; the sum comes from points that violate the desired unit margin. At $m_i = 1$, use a valid subgradient.

Perceptron vs. hinge

For signed score $m = yf(\boldsymbol{x})$:

$$\ell_{\text{perc}} = \max(0, -m), \qquad \ell_{\text{hinge}} = \max(0, 1 - m).$$

- Perceptron loss is zero once $m \geq 0$.
- Hinge loss is zero only when $m \geq 1$.

Difference: SVM also penalizes points that are correctly classified but too close to the boundary.

Soft-margin dual

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Dual form

The soft-margin dual keeps the same objective:

$$\max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^\mathsf{T} \boldsymbol{x}_j.$$

The constraints become:

$$\boxed{0 \leq \alpha_i \leq C}$$

and:

$$\boxed{\sum_i \alpha_i y_i = 0}.$$

The new feature is the upper bound C.

Multiplier upper bound

Soft margin adds $C\xi_i$ to the objective and introduces multipliers $\alpha_i \geq 0$ and $\mu_i \geq 0$ for the two constraints on each slack variable.
The terms involving ξ_i become:

$$\big(C - \alpha_i - \mu_i\big)\xi_i.$$

Stationarity with respect to ξ_i requires:

$$C - \alpha_i - \mu_i = 0 \quad \Longrightarrow \quad \boxed{\alpha_i = C - \mu_i \leq C}.$$

Together with $\alpha_i \geq 0$, we obtain:

$$\boxed{0 \leq \alpha_i \leq C.}$$

Meaning: the same C that prices violations in the primal also caps each training point's multiplier in the dual.

Multiplier and margin cases

The KKT conditions relate α_i to the signed score $m_i = y_i f_i$ and slack ξ_i:

Multiplier	Margin interpretation
$\alpha_i = 0$	$m_i \geq 1$ (often safely outside)
$0 < \alpha_i < C$	$m_i = 1$, $\xi_i = 0$ (on the margin)
$\alpha_i = C$	$m_i \leq 1$ (on or inside margin)

Equality cases may be degenerate, but the table captures the standard geometric interpretation.
Prediction remains $f(\boldsymbol{x}) = \sum_i \alpha_i y_i \boldsymbol{x}_i^\top \boldsymbol{x} + b$: only nonzero multipliers contribute.

Practical SVM

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Feature scaling

SVM depends on distances and inner products, so feature units affect its geometry.
For example, one feature may be around 0.1 while another is around $10{,}000$.

- Standardizing continuous features is often useful.
- Fit scaling on training data only; reuse it for validation and test data.

Common mistake

Scaling must be inside the training or cross-validation pipeline.

Choosing C

Tune C using validation data, for example over:

$$C \in \{0.01, 0.1, 1, 10, 100\}.$$

1. Fit a model for each candidate using training data.
2. Compare validation performance and select C.
3. Evaluate the selected pipeline once on the test set.

Do not use the test set for model selection.

Selecting C with validation

Illustrative results from a binary classification experiment (not measured course data):

C	Training error	Validation error
0.01	15%	18%
1	6%	$\mathbf{9}\%$
100	1%	13%

Choice: $\boxed{C = 1}$ has the lowest validation error among the tested values, even though $C = 100$ fits training data more closely.
Practice: tune C without using the final test set. Standardization must also be learned within each training split.

Class weights

One class may deserve more attention because it is rarer or the consequences of missing it are larger.
Use sample-dependent hinge penalties:

$$\frac{1}{2}\|\boldsymbol{w}\|_2^2 + \sum_i C_{y_i} \max(0, 1 - y_i f_i).$$

Here C_+ and C_- penalize violations by positive and negative observations.
A larger C_+ makes violating a positive-class margin constraint more costly.
Important: class-weighted training is not identical to changing a calibrated probability threshold; its effect must be validated.

Unequal penalties in numbers

Suppose a positive and a negative training observation both have signed score $m = 0.6$.
Both have unweighted hinge loss:

$$\ell_{\text{hinge}} = \max(0, 1 - 0.6) = 0.4.$$

Let $C_+ = 5$ and $C_- = 1$. Their contributions become:

$$\boxed{\text{positive: } 5(0.4) = 2.0, \qquad \text{negative: } 1(0.4) = 0.4.}$$

Meaning: even with the same geometric margin, a positive-class violation can have a larger effect on training.
Use: cost-sensitive or imbalanced classification, subject to validation and appropriate evaluation metrics.

Linear SVM

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

model = make_pipeline(
    StandardScaler(),
    SVC(kernel="linear", C=1.0)
)

model.fit(X_train, y_train)
```

The pipeline ensures that scaling is fitted using the training data seen by the model.
Select C using validation data.

SVM output

A linear SVM produces a score $f(\boldsymbol{x}) = \boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b$.

- $\operatorname{sign}(f(\boldsymbol{x}))$: predicted class for nonzero scores.
- $f(\boldsymbol{x})/\|\boldsymbol{w}\|_2$: signed geometric distance.

The score is **not** automatically a probability. If probabilities are needed, calibrate using held-out or cross-validated data.

A prediction with geometric distance

Suppose a trained SVM has $\boldsymbol{w} = (1, -0.5)^{\mathsf{T}}$ and $b = -0.25$.
Then $\|\boldsymbol{w}\|_2 = \sqrt{1.25} \approx 1.118$.

New point	SVM score f	Prediction	Signed distance $f/\|\boldsymbol{w}\|$
$(2, 1)$	$2 - 0.5 - 0.25 = 1.25$	$+1$	≈ 1.118
$(0, 2)$	$-1 - 0.25 = -1.25$	-1	≈ -1.118

Meaning: these are equally far from the boundary on opposite sides. Neither 1.25 nor -1.25 is a probability.
Use: distances can rank decisions by geometric separation, but they do not directly quantify confidence.

Logistic vs. SVM

Property	Logistic regression	Linear SVM
Loss	Logistic	Hinge
Output	Probability model	Margin score
Correct distant points	Small influence	Zero hinge loss
Main control	Regularization	Margin–violation trade-off

Both produce linear decision boundaries.
The main difference is how the boundary is fitted.

Support-vector dependence

- **Logistic regression:** every finite signed score has nonzero logistic-loss gradient.
- **SVM:** a point strictly outside the margin has zero hinge-loss gradient and $\alpha_i = 0$ at optimum.

Only nonzero dual coefficients contribute to SVM prediction.

Next week: use kernels to obtain nonlinear boundaries.

Week 10 summary

1. **Maximum margin:** choose a separator by geometric distance.
2. **Dual:** support vectors determine the classifier.
3. **Soft margin:** slack and hinge loss handle violations; C sets their cost.

$$\boxed{\text{geometry} \rightarrow \text{optimization} \rightarrow \text{support vectors}}$$

Maximum margin, controlled violations

Questions and discussion